Assignment 3

Game Programming

Name: Fatima Tul Zahra
SAP ID: 59524
Class: BSSE-5
Subject: Game Programming

Multiplayer Game Code Architecture Design

Selected Game: Among Us

1. Game Overview

Among Us is a multiplayer online social deduction game where 4–15 players participate in a match. Most players act as Crewmates and complete tasks, while one or more players act as Impostors who try to eliminate Crewmates without being detected. The game requires real-time networking, player synchronization, task management, voting mechanics, and game state management.

2. High-Level Architecture

Player System
Handles player profiles, movement, tasks, status, and roles.

Networking System
Synchronizes player actions between clients and server.

Matchmaking System
Creates lobbies, manages room joining, and starts matches.

Task System
Assigns and tracks completion of tasks.

Voting & Meeting System
Manages emergency meetings and voting.

Game State Management
Controls game phases such as Lobby, Playing, Meeting, and End Game.

UI System
Displays menus, tasks, voting screens, and notifications.

Audio System
Handles sound effects and game feedback.

3. Code Structure

Main Classes and Responsibilities

GameManager
Controls overall game flow and game states.

PlayerManager
Stores all active players and player information.

Player
Represents an individual player and their behavior.

NetworkManager
Handles communication between server and clients.

LobbyManager
Creates and manages game lobbies.

TaskManager
Assigns and validates player tasks.

VotingManager
Handles meetings and vote calculations.

UIManager
Updates user interface elements.

AudioManager
Controls game audio.

System Communication Diagram

```
                +-----------------+
                |   GameManager   |
                +--------+--------+
                         |
     +-------------------+-------------------+
     |                   |                   |
+----v----+       +------v------+     +------v------+
| Player  |       | Networking  |     |  UIManager  |
| Manager |       |   Manager   |     +-------------+
+----+----+       +------+------+     |                    |
+----v----+       +------v------+
|  Player |       | LobbyManager|
+---------+       +------+------+                          |
                  +------v------+
                  |TaskManager  |
                  +------+------+                          |
                  +------v------+
                  |VotingManager|
                  +-------------+
```

Communication Flow:
Players send actions to the NetworkManager. The NetworkManager validates and forwards data to GameManager. GameManager updates the game state and notifies TaskManager, VotingManager, and UIManager. The updated information is then synchronized across all connected players.

4. Design Decisions

This architecture follows a modular design approach. Each system has a single responsibility, which improves maintainability, scalability, and testing. The GameManager acts as the central controller, reducing dependency between systems. Networking is isolated in a dedicated module, making it easier to support multiplayer synchronization. The architecture is suitable for games like Among Us because it supports real-time communication, lobby management, voting mechanics, and task tracking while keeping the code organized.

Advantages of this architecture include:

- Easy maintenance and debugging
- Better scalability for future features
- Clear separation of responsibilities
- Improved multiplayer synchronization
- Reusable and modular code structure